

Robustness Testing of Foundation Models via Prompt Perturbation

1. Introduction

This project investigates how prompt variations affect the output of a large language model, GPT-3.5, in a sentiment classification task. We aim to evaluate the robustness of GPT-3.5 to different prompt wordings and compare its outputs against an open-source Hugging Face model (CardiffNLP-RoBERTa). The analysis is based on 30 customer reviews of Nike sneakers extracted from Amazon.

2. Methodology

- **Task:** Classify review sentiment into Positive, Neutral, or Negative.
- **Dataset:** 30 customer reviews manually labeled using GPT-3.5.
- **Models:**
 - GPT-3.5 (OpenAI API)
 - CardiffNLP-RoBERTa (Hugging Face Inference API)
- **Prompt Variations:** Six prompts were tested with GPT-3.5 and CardiffNLP:

ID	Description	Prompt
B0	Base prompt	"Classify the sentiment of the following review as Positive, Neutral, or Negative: {review}"
P1	Synonym / Informal tone	"What's the vibe of this review? Choose: Positive, Neutral, or Negative. Review: {review}"
P2	Reversed order	"Review: {review} — What sentiment does this review express: Positive, Neutral, or Negative?"
P3	Question form	"Can you determine if the following review is Positive, Neutral, or Negative? Here it is: {review}"
P4	Formal tone	"Please analyze the sentiment conveyed in the following customer review and classify it as Positive, Neutral, or Negative: {review}"
P5	Instruction wording	"Sentiment classification required. Input: {review}"

3. Evaluation Strategy We used the following evaluation strategies:

1. Output Consistency

- GPT-3.5: 83.3% of reviews had identical predictions across all 6 prompts.
- CardiffNLP: 86.7% consistency.
- Overall GPT ↔ HF Agreement: 71.11%
- Agreement by Prompt:
 - B0 0.800000
 - P1 0.866667
 - P2 0.833333
 - P3 0.833333
 - P4 0.666667
 - P5 0.266667

2. Output Quality Across Variants

- Manual scoring on fragile examples.
- **Fidelity:** GPT-3.5 (avg. 1.92), CardiffNLP (avg. 2.25)
- **Fluency:** Both models scored 3.0 (outputs were short and clear)

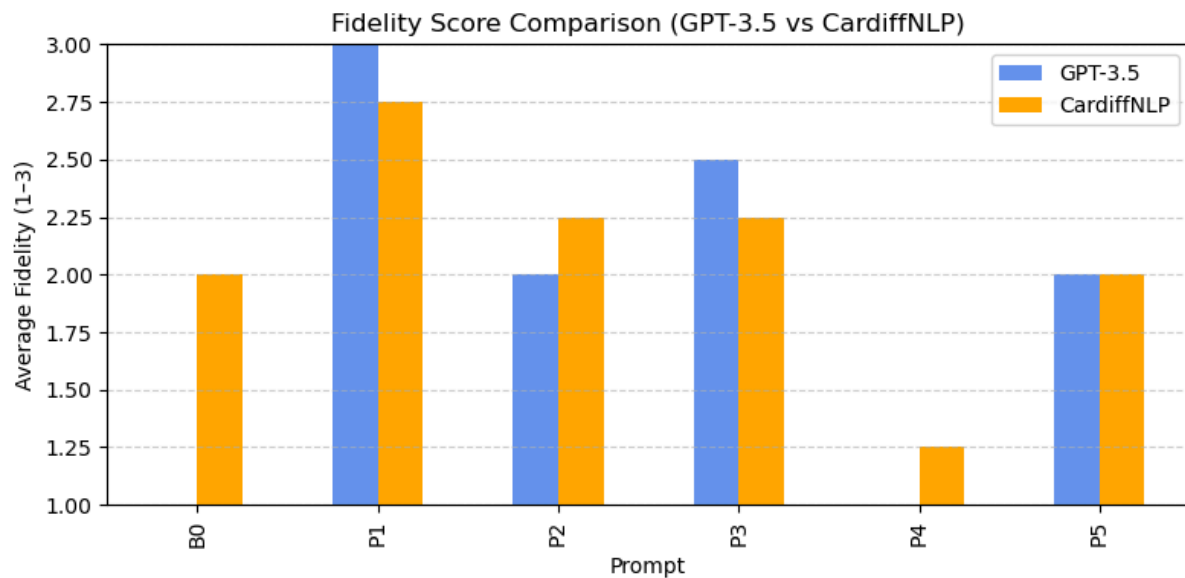


Image 1 — Bar Chart titled "Fidelity Score Comparison (GPT-3.5 vs CardiffNLP)"

3. Statistical Stability

- GPT-3.5: 1.20 unique predictions per review (across prompts)
- CardiffNLP: 1.13 average variance

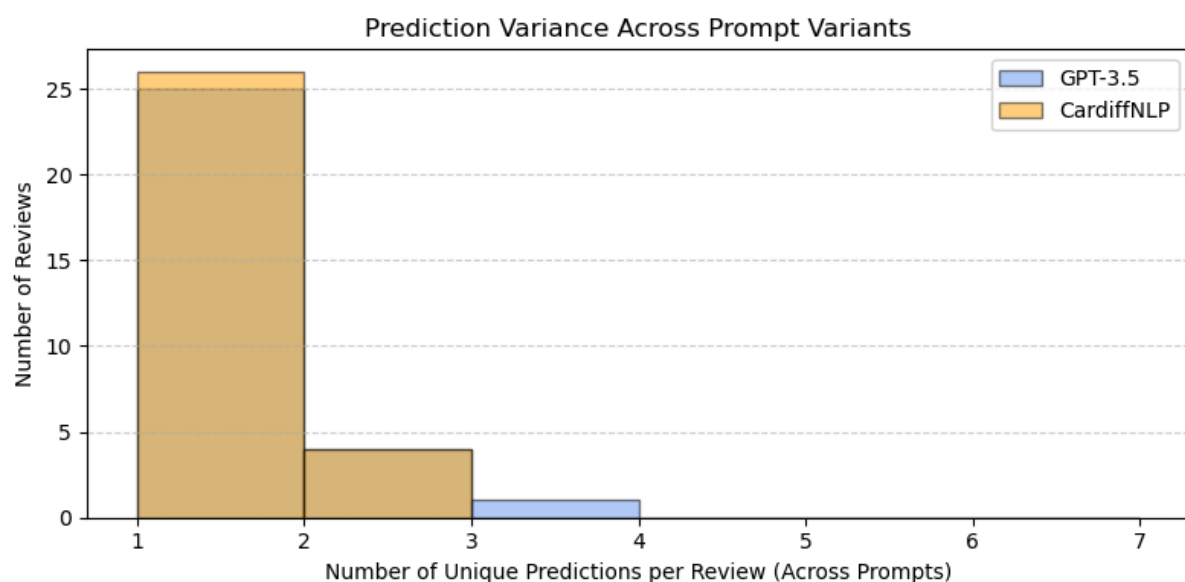


Image 2 — Histogram titled "Prediction Variance Across Prompt Variants"

4. Visualization

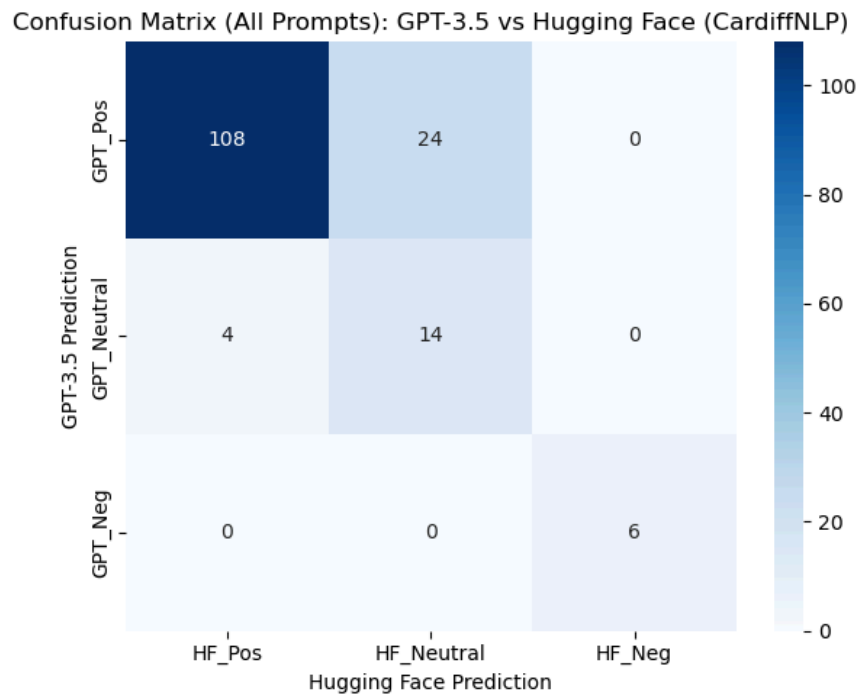


Image 3 — Confusion Matrix: GPT-3.5 vs Hugging Face (CardiffNLP)

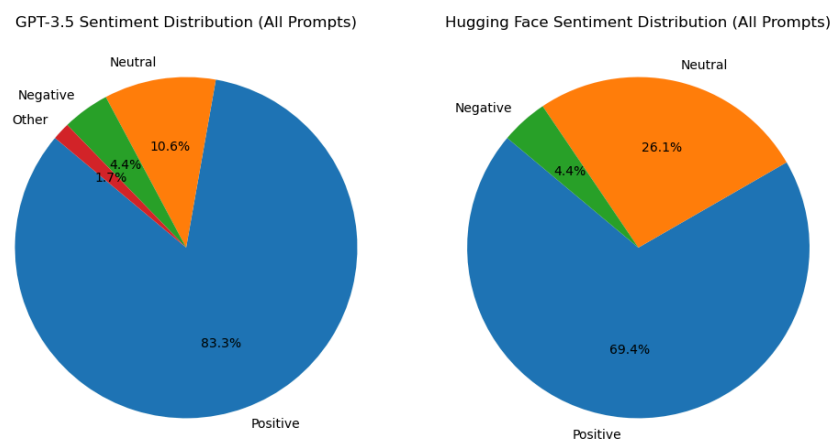


Image 4 — Side-by-side Pie Charts titled "Sentiment Distribution (All Prompts)" for GPT-3.5 and CardiffNLP

4. Results

- **Prompt Robustness (GPT-3.5):**
 - More sensitive to prompt phrasing
 - P1 (informal) and P5 (minimal) performed best in fidelity
- **Model Comparison:**
 - CardiffNLP showed slightly better consistency and less variance
 - GPT-3.5 was more expressive but also more prompt-sensitive

5. Reflection and Reporting

Metric	GPT-3.5	CardiffNLP
Output Consistency	83.3%	86.7%
Average Fidelity (Fragile)	1.92	2.25
Average Fluency (Fragile)	3.00	3.00
Prediction Variance	1.20	1.13

Examples:

- Robust Review: “Absolutely love them, long-time Nike wearer and loyal customer.” — both models: Positive across all prompts
- Fragile Review: “Comfortable and durable, but soles slide easily.” — GPT and CardiffNLP: Neutral/Negative/Positive depending on prompt

Reflection:

- Most fragile prompts: B0, P4 (more formal/ambiguous phrasing)
- Most robust prompts: P1, P5 (informal, minimal prompts)
- Model Robustness: CardiffNLP was more stable overall, while GPT-3.5 was more expressive but required careful prompt design
- Implications: For production, GPT-3.5 must be stress-tested across prompts; CardiffNLP is safer for fixed-format tasks but less adaptable

6. Conclusion

Prompt engineering has a significant impact on model output, even for straightforward sentiment tasks. GPT-3.5 demonstrated variability under prompt perturbation, emphasizing the need for robustness evaluations. Hugging Face models like CardiffNLP offer reliable alternatives for direct classification but are less flexible in prompt-driven setups.

This study reinforces that both model selection and prompt design play critical roles in NLP system performance and trustworthiness.